- Saudi Arabia has plans to establish a civil nuclear power industry.
- The country in 2013 projected 17 GWe of nuclear capacity by 2032, but that target was abandoned two years later.
- At present, virtually all of the country's electricity is generated by burning fossil fuels.

Privacy - Terms

Electricity sector

Total generation (in 2023): 441 TWh

Generation mix: natural gas 257 TWh (58%); oil 182 TWh (41%); wind 1.6 TWh.

Import/export balance: 0.3 TWh imports, 0.4 TWh exports

Total consumption: 318 TWh

Per capita consumption: c. 9400 kWh in 2023.

Source: International Energy Agency and The World Bank. Data for year 2023.

Saudi Arabia's population has grown from 4 million in 1960 to over 35 million in 2024. It consumes over one-quarter of its oil production, and while energy demand is projected to increase substantially, oil production is not.

Generating capacity was 83.0 GWe in 2021, up from 25.8 GWe in 2000. Saudi Arabia is unique in the region in having 60 Hz grid frequency, which severely limits the potential for grid interconnections.

The country has had ambitious plans for building nuclear and renewable energy capacity, but in 2019 only 0.1% of total generation came from solar PV, with the rest being produced by oil and gas.

Plans for renewables

The Saudi government in 2012 adopted a proposal by the King Abdullah City for Atomic and Renewable Energy (KA-CARE) to add 23.9 GWe of renewable capacity by 2020 and 54 GWe by 2032, comprising 16 GWe solar PV, 25 GWe concentrated solar power, 9 GWe wind, 3 GWe waste-to-energy, and 1 GWe geothermal. This plan was later pushed back to 2040. In April 2016 the government launched its Vision 2030 initiative that included plans under the National Renewable Energy Program (NREP) for 9.5 GW

of renewable energy by 2023 with an interim target of 3.45 GWe by 2020. The Vision 2030 targets were increased in January 2019 by the country's Renewable Energy Project Development Office (REPDO) to 27.3 GWe in 2023 and 58.7 GWe in 2030, comprising 40 GWe solar PV, 16 GWe wind and 2.7 GWe other renewable sources.

In March 2018 Crown Prince Mohammed bin Salman signed a memorandum of understanding (MoU) with Japan's SoftBank to deliver 150-200 GW of solar capacity by 2030 as part of a $200 billion programme. The project was aborted about six months later.

Earlier, in October 2017, Crown Prince Mohammad bin Salman announced the $500 billion Neom project, a 2.7 million ha new city on the Red Sea in the northwest of the country, to be fully powered by renewable energy. In March 2022, Neom announced the launch of subsidiary company Enowa, which is responsible for developing Neom's energy and water systems.

In January 2021 energy minister Prince Abdulaziz bin Salman said the country is committed to becoming carbon neutral and that it aimed to produce 50% of its electricity from renewables by 2030, with the remaining 50% supplied by natural gas.

Desalination

The Ministry of Water & Electricity (MOWE), which had been broadly responsible for power and desalination in the country, was broken up in May 2016, with the water portfolio going to the Ministry of Environment, Water and Agriculture, and electricity to the Ministry of Energy, Industry and Mineral Resources.

The Saline Water Conversion Corporation (SWCC) operates about 7.5 million m^3/d of desalinated water capacity. Coupling desalination plants with power generation so as to use reject heat reduces energy requirements for desalination by about half. Hence dual-purpose or hybrid plants are favoured, as independent water and power production (IWPP) facilities. The country continues to install huge desalination capacity, much of it thermal multi-stage flash distillation (MSF) and multiple-effect distillation (MED), but a lot is reverse osmosis (RO), driven by electricity.

The Shuaibah/Shoaiba plant on the Red Sea coast provides 0.9 million m^3/d capacity; Yanbu and nearby Rabigh 3 on the Red Sea supplying the Medina region have another 1.5 million m^3/d. On the

Gulf coast Ras Al Khair has over 1 million m^3/d supplying Riyadh and Marafiq/Jubail provides about 0.8 million m^3/d.

In January 2020 UK-based Solar Water was contracted to build the first 'solar dome' concentrated solar power desalination plants in Neom, with the first plant expected to be completed later in 2020.

Nuclear power industry

In December 2006 the six member states of the Gulf Cooperation Council (GCC) – Kuwait, Saudi Arabia, Bahrain, United Arab Emirates (UAE), Qatar and Oman – announced that the Council was commissioning a study on the peaceful use of nuclear energy. France agreed to work with them on this, and Iran pledged assistance with nuclear technology.

In February 2007 the six states agreed with the International Atomic Energy Agency (IAEA) to cooperate on a feasibility study for a regional nuclear power and desalination programme, with Saudi Arabia leading the investigation.

In August 2009 the Saudi government announced that it was considering a nuclear power programme on its own, and in April 2010 a royal decree said: "The development of atomic energy is essential to meet the Kingdom's growing requirements for energy to generate electricity, produce desalinated water and reduce reliance on depleting hydrocarbon resources." The King Abdullah City for Atomic and Renewable Energy (KA-CARE) was set up in Riyadh to advance this agenda as an alternative to oil and to be the competent agency for treaties on nuclear energy signed by the kingdom. It is also responsible for supervising works related to nuclear energy and radioactive waste projects.

In June 2010 it appointed the Finland- and Swiss-based Poyry consultancy firm to help define "high-level strategy in the area of nuclear and renewable energy applications" with desalination. In November 2011 it appointed WorleyParsons to conduct site surveys and regional analysis to identify potential sites, to select candidate sites then compare and rank them, and to develop technical specifications for a planned tender for the next stage of the Saudi nuclear power project. Three sites were shortlisted as of September 2013: Jubail on the Gulf; and Tabuk and Jizan on the Red Sea.

Earlier in June 2011 the coordinator of scientific collaboration at KA-CARE said that it planned to construct 16 nuclear power reactors over the next 20 years at a cost of more than 300 billion riyals ($80

billion). These would generate about 20% of Saudi Arabia's electricity. Smaller reactors such as Argentina's CAREM were envisaged for desalination. An April 2013 timeline showed nuclear construction starting in 2016.

In April 2013 KA-CARE projected 17 GWe of nuclear capacity by 2032 of total 123 GWe, with 16 GWe solar PV, 25 GWe solar CSP (to provide for heat storage), and 4 GWe from geothermal, wind and waste. About half the capacity in 2032 would still be hydrocarbon, with one-third solar following investment in that of some $108 billion. In addition 9 GWe of wind capacity would be used for desalination. In January 2015 the nuclear target date was moved to 2040.

In September 2013 both GE Hitachi Nuclear Energy and Toshiba/Westinghouse signed contracts with Exelon Nuclear Partners (ENP), a division of Exelon Generation, to pursue reactor construction deals with KA-CARE. GEH proposed its ABWR and ESBWR, while Toshiba/Westinghouse proposed the AP1000 and Toshiba's ABWR. Areva and EdF signed a number of agreements with Saudi companies and universities, and EdF signed an agreement with Saudi Arabia's Global Energy Holding Company (GEHC) for the creation of a joint venture whose first task would be to carry out feasibility studies for an EPR reactor in the country.

In January 2015 the government said that its target for 17 GWe of nuclear capacity would be more like 2040. This effectively meant that immediate plans were scaled back, and in 2017 KA-CARE announced that it was soliciting proposals for 2.9 GWe nuclear capacity, from South Korea, China, Russia and Japan. In November 2018 KA-CARE awarded a contract to Worley Parsons to provide consultancy services for the Saudi National Atomic Energy Project. This covers project governance, resource management, project services, training and compliance across the full scope of large plants, small modular reactors (SMR) and the nuclear fuel cycle.

Earlier in March 2015 the Korea Atomic Energy Research Institute (KAERI) signed an agreement with KA-CARE to assess the potential for building at least two South Korean SMART reactors in the country, and possibly more. The pre-project engineering study was expected to take three years, and in late 2016 was "progressing for the FOAK plant construction". In September 2015 further contracts were signed, which aim at building a partnership to establish knowledge infrastructure in SMART technology fields, such as designing and building the reactors and maintaining their mechanical and safety features. The two nations invested $130

million from 2015 to November 2017 to complete pre-SMART construction engineering.

SMART is designed for electricity generation (up to 100 MWe) as well as thermal applications, such as seawater desalination, with a 60-year design operating lifetime and three-year refuelling cycle. The cost of building the first SMART unit in Saudi Arabia is estimated at $1 billion. The agreement is seen by South Korea as opening opportunities for major involvement in Saudi nuclear power plans, and it also calls for the commercialization and promotion of the SMART reactor to third countries. KAERI has designed an integrated desalination plant based on the SMART reactor to produce 40,000 m^3/day of water and 90 MWe of power at less than the cost of gas turbine.

Also in March 2015, the state-owned INVAP (Investigacion Aplicada) from Argentina and state-owned Saudi technology innovation company Taqnia set up a joint venture company, Invania, to develop nuclear technology for Saudi Arabia's nuclear power program, apparently focusing on small reactors such as CAREM (100 MWt, 27 MWe) for desalination. Taqnia is the technology arm of the Public Investment Fund.

In January 2016 KA-CARE signed an agreement with China Nuclear Engineering Corporation (CNEC) to build a high-temperature reactor (HTR) in the country, based on the HTR-PM now under construction in China by CNEC. A further cooperation agreement to this end, including localization of the supply chain and undertaking a feasibility study, was signed in March 2017. In May 2017 a joint working group commenced a formal feasibility study for the project, with a view to submitting it to the government later in 2017. In August 2017 China Nuclear Engineering & Construction Group (CNEC) and Saudi Technology Development Corporation signed an agreement for a feasibility study on using high temperature reactors for seawater desalination. The IAEA also reports a feasibility study on HTRs providing heat for the petro-chemical industry in Saudi Arabia.

In July 2017 the cabinet approved the establishment of the Saudi National Atomic Energy Project (SNAEP), and new financial and administrative regulations for KA-CARE.

In January 2019 the IAEA delivered the final report of its integrated nuclear infrastructure review (INIR) mission in Saudi Arabia, concluding that "significant progress" had been made, including the establishment of a legislative framework and development of nuclear infrastructure.

In February 2022 Saudi Arabia confirmed the establishment of the Nuclear Holding Company, which will act as the country's nuclear developer. The entity's formation was originally mentioned in 2013.

In September 2023 the energy minister reiterated the Kingdom's intention to build a nuclear power plant, including plans to rescind the small quantities protocol and switch to a comprehensive safeguards agreement with the International Atomic Energy Agency (IAEA).

International agreements

A nuclear cooperation agreement with France in early 2011 seemed likely to advance French interests in the country's plans. In June 2015 France signed an agreement to undertake a feasibility study for building two EPR nuclear power reactors. Additional agreements were signed on nuclear safety training as well as on waste disposal.

A mid-2011 nuclear cooperation agreement with Argentina was evidently related to smaller plants for desalination and the subsequent Invania joint venture.

A November 2011 agreement with South Korea called for cooperation in nuclear R&D, including building nuclear power plants and research reactors, as well as training, safety and waste management. In June 2013 Kepco offered support for the localization of nuclear technology, along with joint research and development of nuclear technologies if Saudi Arabia purchases South Korean reactors. In September 2015 further contracts were signed, which aim at building a partnership to establish knowledge infrastructure in SMART technology fields (see March 2015 SMART agreement above).

A January 2012 agreement with China relates to nuclear plant development and maintenance, research reactors, and the provision of fabricated nuclear fuel. A further agreement with CNNC was signed in August 2014, and in August 2016 KA-CARE signed an agreement with CNNC for human resource development.

A June 2015 agreement with Rosatom provided for cooperation in the field of nuclear energy, including: the design, construction, operation and decommissioning of nuclear power and research reactors, including desalination plants and particle accelerators; the provision of nuclear fuel cycle services, including nuclear power plants and research reactors; the management of used nuclear fuel

and radioactive waste management; the production of radioisotopes and their application in industry, medicine and agriculture; and the education and training of specialists in the field of nuclear energy. A further programme of cooperation was signed in October 2017 between KA-CARE and Rosatom, focused on small and medium reactors, and on building a new research reactor. In July 2019 it was reported that Rosatom proposed a feasibility study on building a VVER-600 reactor for KA-CARE initially, with a view to building VVER-1200 units eventually.

In October 2015 KA-CARE signed a nuclear cooperation agreement with Hungary. In October 2016 it signed a nuclear cooperation agreement with Kazakhstan, focused on fuel supply.

In March 2017 an agreement between KA-CARE and Jordan Atomic Energy Commission (JAEC) was signed for a feasibility study on the construction of two small modular reactors (SMRs) in Jordan for the production of electricity and desalinated water. No particular technology was mentioned.

KA-CARE earlier said it was negotiating with the Czech Republic, UK and the USA regarding "further cooperation". A full nuclear cooperation agreement with the USA is generally seen as vital to proceeding with Saudi nuclear power plans.

In November 2025 the US and Saudi Arabian governments announced the completion of negotiations on a civil nuclear cooperation framework, which establishes the basis for a long-term partnership and potential US reactor exports.

Fuel cycle

There have been media reports speculating on KA-CARE intentions for uranium enrichment, without confirmation or plausibility. The speculation is based on a possible nuclear agreement with the USA, and relates to regional politics.

In March and August 2017 China National Nuclear Corporation (CNNC) and the Saudi Geological Survey signed agreements on cooperation on the exploration of uranium. CNNC said it would explore nine potential areas for uranium resources in Saudi Arabia over the next two years. It has also been exploring in Jordan.

In March 2019 KA-CARE launched a programme in conjunction with the Jordan Atomic Energy Commission (JAEC) and the Jordan Uranium Mining Company (JUMCO) to develop Saudi expertise in uranium exploration and mining. This follows a March 2017

agreement between JAEC and KA-CARE covering uranium exploration and mining in central Jordan.

Research & development

A 30 kWt low power research reactor (LPRR) is under construction at King Abdulaziz City for Science & Technology (KACST) in Riyadh by Argentina's INVAP. The project was officially launched in November 2018, when construction was reported to be well under way.

Regulation, safety and non-proliferation

The Saudi Arabian Atomic Regulatory Authority (SAARA) was set up to commence activities early in 2014. In May 2014 KA-CARE signed an agreement with the Finnish Radiation and Nuclear Safety Authority (STUK) to assist in this by recruiting and training personnel and establishing safety standards. In November 2016 KA-CARE signed an agreement with South Korea's Nuclear Safety and Security Commission (NSSC) to promote cooperation in "regulating nuclear safety, safeguards and physical protection, radiation protection and relevant research, as well as development in a manner to serve atomic energy programmes in the Kingdom of Saudi Arabia," according to KA-CARE. NSSC said that "the platform of cooperation" was "expected to play an imperative role in facilitating bilateral cooperations in the region."

In May 2021 Saudi Arabia's Nuclear and Radiological Regulatory Commission (NRRC) and the UAE's Federal Authority for Nuclear Regulation (FANR) agreed to cooperate in nuclear and radiation regulatory matters. The two regulators would form topical working groups to share best practice. This follows the signature in 2019 of a bilateral accord on cooperation in nuclear energy between the two countries.

Non-proliferation

Saudi Arabia has had a safeguards agreement in force with the IAEA since 2009, but no Additional Protocol. Argentina's envoy to the IAEA has said that further safeguards arrangements will be needed before the research reactor it is building at KACST can be fuelled.

Notes & references

Korea Atomic Energy Research Institute, KAERI and K.A.CARE signed SMART PPE Agreement (3 September 2015)